

Math 526, S06
Exam 1

Name:
SSN: xxx-xx-

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.
- No graphical calculator is allowed in the exam.

Problem	Points	Score
1	20	
2	20	
3	20	
4	20	
5	20	
Total	100	

Good Luck !

Problem 1. (20 points) Let $\mathbf{A} = \begin{bmatrix} 0 & 2 & 4 \\ -3 & 1 & -3 \\ 3 & 5 & -1 \\ -1 & 2 & 5 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 2 & -1 & 3 \\ -2 & 4 & -2 \end{bmatrix}$.

a) Which one of the products $\mathbf{AB}^t$ and $\mathbf{A}^t\mathbf{B}$ is possible? Explain.

Solution: $\mathbf{AB}^t$ is possible since the dimension of $\mathbf{A}$ is 4×3 and the dimension of $\mathbf{B}^t$ is 3×2 . The product is a 4×2 matrix.

b) Compute the product from a) that is possible.

Solution:

$$\begin{aligned} \mathbf{AB}^t &= \begin{bmatrix} 0 & 2 & 4 \\ -3 & 1 & -3 \\ 3 & 5 & -1 \\ -1 & 2 & 5 \end{bmatrix} \cdot \begin{bmatrix} 2 & -2 \\ -1 & 4 \\ 3 & -2 \end{bmatrix} \\ &= \begin{bmatrix} 10 & 0 \\ -16 & 16 \\ -2 & 16 \\ 11 & 0 \end{bmatrix} \end{aligned}$$

Problem 2. (20 points)

a) Find two 2×2 matrices $\mathbf{A}$ and $\mathbf{B}$ such that $\mathbf{AB} = \mathbf{0}_{2 \times 2}$, but $\mathbf{A} \neq \mathbf{0}_{2 \times 2}$ and $\mathbf{B} \neq \mathbf{0}_{2 \times 2}$.

Solution: There are many solutions. For example:

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$$

or

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$$

b) Consider the 2×3 matrices

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 4a_{11} - a_{21} & 4a_{12} - a_{22} & 4a_{13} - a_{23} \\ 5a_{11} - 3a_{21} & 5a_{12} - 3a_{22} & 5a_{13} - 3a_{23} \\ -3a_{11} + a_{21} & -3a_{12} + a_{22} & -3a_{13} + a_{23} \end{bmatrix}.$$

Find a 3×2 matrix $\mathbf{B}$ so that $\mathbf{A}$ multiplied by $\mathbf{B}$ equals $\mathbf{C}$. From which side should you multiply $\mathbf{A}$ by $\mathbf{B}$?

Solution: We have $\mathbf{BA}$ where

$$\mathbf{B} = \begin{bmatrix} 4 & -1 \\ 5 & -3 \\ -3 & 1 \end{bmatrix}$$

Problem 3. (20 points) Let $\mathbf{A} = \begin{bmatrix} -1 & -2 & -1 \\ -1 & 1 & -2 \\ -2 & 4 & -8 \end{bmatrix}$.

a) Find the $\mathbf{L}-\mathbf{U}-\mathbf{P}$ factorization ($\mathbf{LU} = \mathbf{PA}$) of $\mathbf{A}$ using the strategy of **partial pivoting**.

Solution:

$$\begin{array}{ccc} \left[\begin{array}{c} \\ \\ \end{array} \right] & \left[\begin{array}{ccc} -1 & -2 & -1 \\ -1 & 1 & -2 \\ -2 & 4 & -8 \end{array} \right] & \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right] \\ \left[\begin{array}{c} \\ \\ \end{array} \right] & \left[\begin{array}{ccc} -2 & 4 & -8 \\ -1 & 1 & -2 \\ -1 & -2 & -1 \end{array} \right] & \left[\begin{array}{ccc} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{array} \right] \\ \left[\begin{array}{c} 1/2 \\ 1/2 \\ \end{array} \right] & \left[\begin{array}{ccc} -2 & 4 & -8 \\ 0 & -1 & 2 \\ 0 & -4 & 3 \end{array} \right] & \left[\begin{array}{ccc} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{array} \right] \\ \left[\begin{array}{c} 1/2 \\ 1/2 \\ \end{array} \right] & \left[\begin{array}{ccc} -2 & 4 & -8 \\ 0 & -4 & 3 \\ 0 & -1 & 2 \end{array} \right] & \left[\begin{array}{ccc} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{array} \right] \\ \left[\begin{array}{cc} 1/2 & 1/4 \end{array} \right] & \left[\begin{array}{ccc} -2 & 4 & -8 \\ 0 & -4 & 3 \\ 0 & 0 & 5/4 \end{array} \right] & \left[\begin{array}{ccc} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{array} \right] \end{array}$$

So we have

$$\mathbf{L} = \begin{bmatrix} 1 & & & \\ 1/2 & 1 & & \\ 1/2 & 1/4 & 1 & \end{bmatrix}, \quad \mathbf{U} = \begin{bmatrix} -2 & 4 & -8 \\ 0 & -4 & 3 \\ 0 & 0 & 5/4 \end{bmatrix}, \quad \mathbf{P} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

b) Find the corresponding $\mathbf{L}-\mathbf{D}-\mathbf{U}-\mathbf{P}$ factorization ($\mathbf{LDU} = \mathbf{PA}$) of $\mathbf{A}$ using the result from a).

Solution: $\mathbf{L}$ and $\mathbf{P}$ have no change and

$$\mathbf{D} = \begin{bmatrix} -2 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & 5/4 \end{bmatrix}, \quad \mathbf{U} = \begin{bmatrix} 1 & -2 & 4 \\ 0 & 1 & -3/4 \\ 0 & 0 & 1 \end{bmatrix}.$$

Problem 4. (20 points) Let $\mathbf{A} = \begin{bmatrix} 2 & 0 & 0 \\ 3 & 1 & 0 \\ -4 & 2 & 2 \end{bmatrix}$.

a) Explain why $\mathbf{A}$ is invertible.

Solution: Since $\mathbf{A}$ is a lower triangular matrix and there are no zero's in the diagonal of $\mathbf{A}$, $\mathbf{A}$ is invertible.

b) Find the inverse matrix of $\mathbf{A}$.

Solution: The first column of $\mathbf{A}^{-1}$ is computed by solving

$$\mathbf{A}\mathbf{x} = \mathbf{e}_1$$

i.e.,

$$\begin{bmatrix} 2 & 0 & 0 \\ 3 & 1 & 0 \\ -4 & 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

We get

$$x_1 = 1/2, \quad x_2 = -3/2, \quad x_3 = 5/2.$$

The second column of $\mathbf{A}^{-1}$ is computed by solving

$$\begin{bmatrix} 2 & 0 & 0 \\ 3 & 1 & 0 \\ -4 & 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

We get

$$x_1 = 0, \quad x_2 = 1, \quad x_3 = -1.$$

The third column of $\mathbf{A}^{-1}$ is computed by solving

$$\begin{bmatrix} 2 & 0 & 0 \\ 3 & 1 & 0 \\ -4 & 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

We get

$$x_1 = 0, \quad x_2 = 0, \quad x_3 = 1/2.$$

So

$$\mathbf{A}^{-1} = \begin{bmatrix} 1/2 & 0 & 0 \\ -3/2 & 1 & 0 \\ 5/2 & -1 & 1/2 \end{bmatrix}$$

Problem 5. (20 points)

a) How many flops does it take to compute the product of a tridiagonal $n \times n$ matrix and a column n -vector if multiplication by zero is not executed? Explain.

Solution: $5n - 4$

b) Let $\mathbf{u} = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$. Is the matrix $\mathbf{I}_{4 \times 4} - \mathbf{u}\mathbf{v}^t$ nonsingular? Explain.

Solution: It is nonsingular since

$$\mathbf{u}^t \mathbf{v} = \begin{bmatrix} 1 & -1 & 1 & -1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = -2 \neq 1$$

(See **Lemma 8.7** in the Lecture Note)